Lars Larsen

Embedded C++ and FPGA Engineer, real-time video

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SUMMARY

Embedded engineer with 8 years of C/C++ firmware on Zynq-7000 (bare-metal and FreeRTOS), STM32F4 and RP2040/RP2350, plus FPGA video design on iCE40 and Zynq. Primary author of the shared firmware framework and drivers used in TBC2, Memory Palace, Chromagnon and Videomancer.

EXPERIENCE

Founder & Principal Engineer, Firmware / FPGA, LZX Industries

Portland, OR · 2011 – Present

Maker of video synthesizers and video processing instruments (DBA 2011, LLC 2017). Lead designer of every product, in a three-person core team with partners on firmware/HDL and board layout.

- Led more than 100 hardware products from concept to production across six Eurorack series, standalone instruments, DIY kits, cases and power. About 240 production circuit board designs and 140 metal front panels. $6.4M in direct sales (2018 to 2026), plus a dealer channel.
- Videomancer (shipping since late 2025): designed a standalone FPGA video synthesizer around an iCE40 HX4K and RP2040, with ADV7611/ADV7513 HDMI in and out, ADV7181C and ADA4412-3 analog input, and an ADV7393 SD/ED/HD encoder. Users load effects as Ed25519-signed packages. Released 69 public firmware release candidates and an open GPL-3.0 SDK.
- Embedded C++ since 2018: primary firmware author for TBC2 and Memory Palace (Zynq-7010, bare-metal and FreeRTOS), Scrolls (STM32F411), and Chromagnon and Videomancer (RP2040). Built the shared application engine (state machines, UI, presets, modulation, MIDI, USB) over three generations; the current C++20 version uses no exceptions, RTTI, STL containers or heap.
- ESG3 Encoder & Sync Generator (Gen3, 2022): pipelined the sync timing path on an iCE40 Ultra so SD, HD and tri-level sync met timing at HD pixel clocks with fixed latency and accurate phase. Added VCXO PLL genlock. The FPGA12 board it runs on is shared by the Gen3 FPGA modules and power/sync entry products.
- Wrote bare-metal drivers for video decoders, encoders and HDMI chips (ADV7181C, ADV7393, ADV7611, ADV7513, THS7353, ADA4412-3), DACs and ADCs (MCP4728, ADS1115, MCP3208), I/O expanders and an audio codec (TLV320AIC3104), with host-side unit tests using mocks and a PC simulator.
- Chromagnon: moved the design from a Zynq SoM to an iCE40HX4K and RP2040. The iCE40 logic recovers timing from embedded SAV/EAV codes, converts YCbCr to RGB and genlocks 15 SD/ED/HD formats up to 74.25 MHz. Brought up passthrough, sync output and feedback on all 15 formats.
- Chromagnon (2026): wrote a Python model of the instrument that matches the iCE40 RTL and GHDL bit for bit, with nine block-level test runners (all 76 injected faults caught in simulation), Rigol MSO5000 control over SCPI, and 88 reference-vector tests for the browser version.
- Software: led the Shopify Hydrogen/React storefront since 2023 (about 58k lines, Playwright tests). In 2026, using AI coding agents: a Python operations CLI (about 1,000 commands, 3,500 tests) linking Shopify, InvenTree, ERPNext, JLCPCB and UPS; a Rust/Tauri firmware updater; self-hosted Forgejo CI and Authentik SSO.

Operations Manager, Darkplace Manufacturing

Portland, OR · 2015 – 2017

Manufacturing arm for the LZX Industries product line.

- Managed operations, purchasing and customer communication for a manufacturing team of 10+ people building the LZX product line, while continuing as its product designer.

Earlier

- Multimedia Programmer, University of North Texas, Denton TX, 2008–2010 and 2011–2015
- Lead Service Technician, Switched On (synthesizer sales and repair), Austin TX, 2010
- New Media Developer, FUNimation Entertainment, Flower Mound TX, 2007–2008

- Multimedia Developer, TargetScope, Plano TX, 2006–2007

SELECTED PROJECTS

- **Videomancer SDK (GPL-3.0)**: iCE40 video platform, program ABI, signed package format and GHDL image tester. About 190 C++ and 350 VHDL unit tests. github.com/lzxindustries/videomancer-sdk
- **Chromagnon web simulator**: Browser version of the Chromagnon instrument model (WebGL), checked against reference vectors. lzxindustries.net/instruments/chromagnon/tools/simulator
- **Videomancer community programs**: Open VHDL video effects written by outside contributors using the SDK. github.com/lzxindustries/videomancer-community-programs

SKILLS

Firmware & FPGA	C/C++ (C++17/20, no heap), Zynq-7000 bare-metal and FreeRTOS, STM32F4, RP2040/RP2350, AVR, TinyUSB, Ed25519 signing, VHDL, Verilog, GHDL, VUnit, Yosys/nextpnr (iCE40), Vivado (Zynq-7000)
Electronics	Analog, mixed-signal and video (SD to 1080p, composite, S-Video, YPbPr, RGB, HDMI, genlock), discrete high-speed analog, power entry and regulation, ngspice/SPICE, IBIS, DipTrace, Eagle, KiCad, DFM, BOM costing, board bring-up, oscilloscope automation
Software	Python (NumPy, OpenCV, PySide6), TypeScript/React (Shopify Hydrogen/Remix, Vitest, Playwright), Docker, Forgejo CI, Linux servers, AI coding tools (Claude Code, Cursor)
Product & operations	Concept to production, PLM (MRPeasy, InvenTree), ERP, supplier and contract manufacturer management, pricing, documentation, open-source hardware
Mechanical	Inventor, FreeCAD, build123d, sheet-metal enclosures, injection-molded knobs, front panel graphics, Blender rendering

EDUCATION

- B.A., Radio, Television & Film, University of North Texas, 2006